ARDUINO-BASED ULTRASONIC RADAR SYSTEM

WITH REAL-TIME OBJECT DETECTION AND ANGULAR POSITION ESTIMATION

Intelligent Sensing • Real-Time Tracking • Smart Detection



INTRODUCTION

Object detection is a fundamentally important aspect of modern radar sensing systems. Combining object detection with ultrasonic sensing technology—such as an embedded ultrasonic array—enables efficient processing to prevent collisions and provides a reliable foundation for compact radar systems.

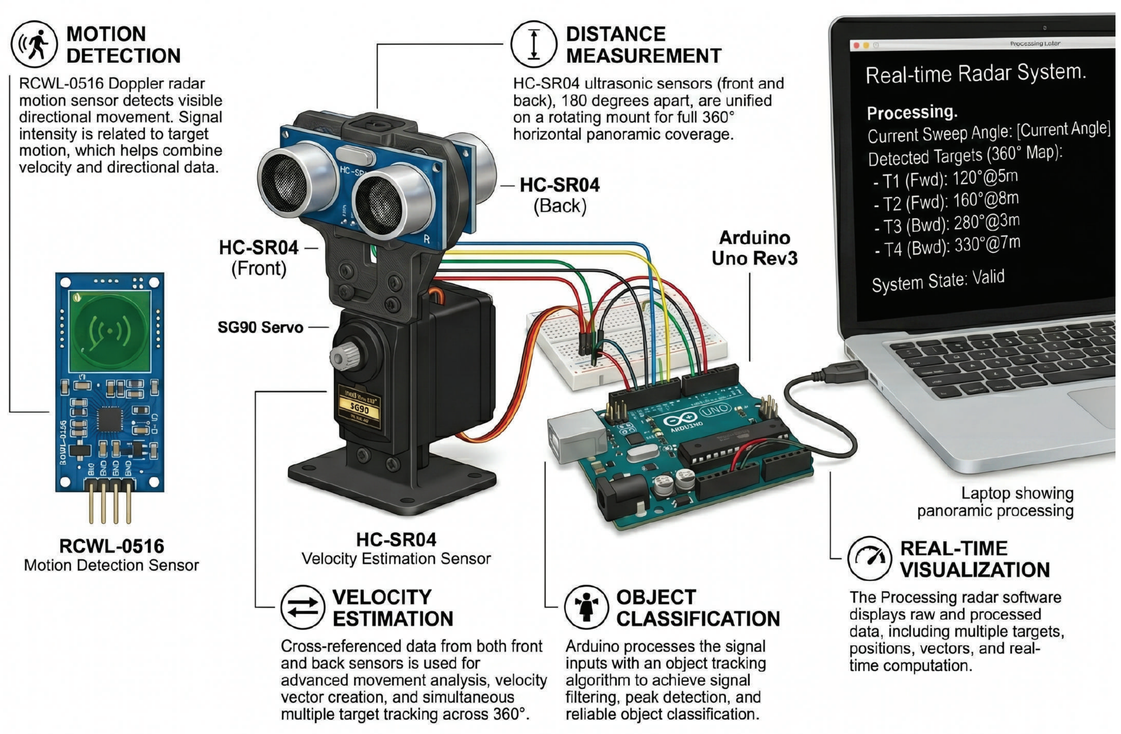
Ultrasonic sensing technology is widely used in low-cost radar mapping applications. Integrating ultrasonic sensing with microcontroller technology allows for real-time computational comparisons with traditional radar systems. The implementation of this ultrasonic embedded system serves both educational purposes and practical applications, demonstrating robust performance for proximity and health monitoring implementations.

Embedded systems utilizing multi-sensor setups excel at target detection and localization. They serve as a cornerstone in educational environments, bridging the gap between theoretical signal processing and hands-on laboratory practices.

OBJECTIVES

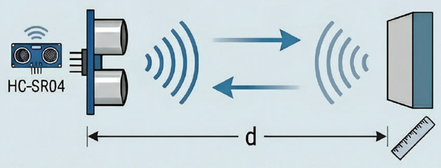
- Detect objects in real time
- Measure object distance accurately
- Estimate the angular position of targets
- Generate real-time radar visualizations
- Integrate microwave-based motion detection
- Build a low-cost, portable radar platform
- Demonstrate embedded sensing and signal processing principles

SYSTEM OVERVIEW & DIAGRAM LABELS



DISTANCE CALCULATION FORMULA

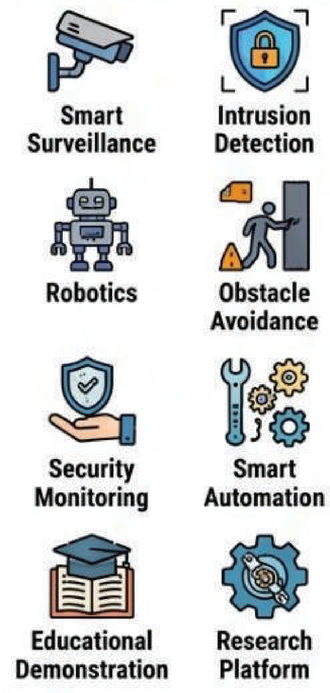
The HC-SR04 determines object distance by measuring the time-of-flight of an ultrasonic pulse. Since the pulse travels to the target and returns to the sensor, the measured travel path is twice the actual object distance.



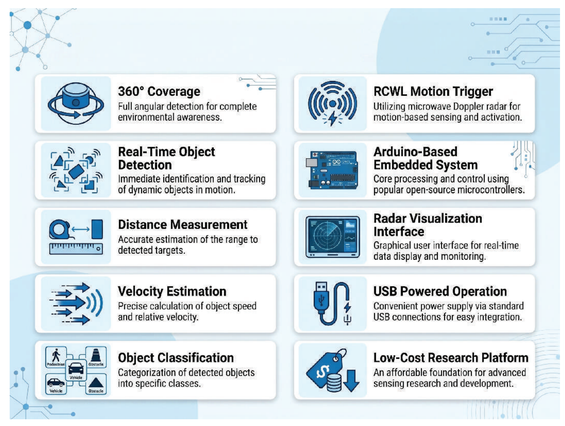
$$d = \frac{v \times t}{2}$$

where:
d = distance to object
v = speed of sound in air
t = measured echo time

APPLICATIONS



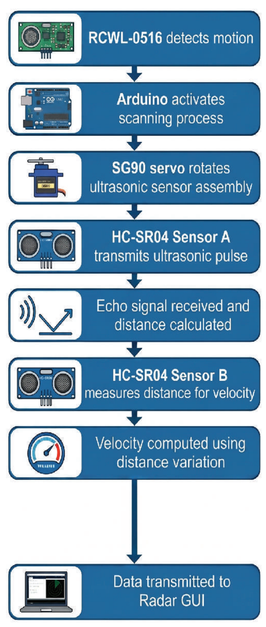
KEY FEATURES



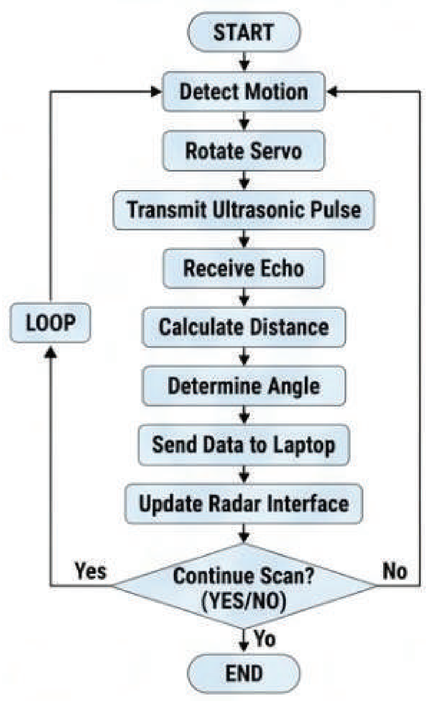
HARDWARE COMPONENTS

Component	Function
	Arduino Uno, Serves as the central microcontroller processing unit to calculate and process incoming ultrasonic data.
	HC-SR04 Sensors Emits high-frequency sound waves and receives echo signals to track obstacles.
	SG90 Servo Motor, Rotates the sensor array across a designated sweep angle to capture angular data.
	RCWL-0516 Microwave Sensor Detects human or object motion within the vicinity using Doppler radar technology.
	Breadboard Provides a solderless platform for organizing and prototyping the electronic circuit.
	Jumper WiresConnects the electronic components, sensors, and microcontroller.
	Laptop Processes serial data and hosts the graphical user interface for radar visualization.
	USB Cable Connects the Arduino to a laptop for data transfer (serial communication) and power.

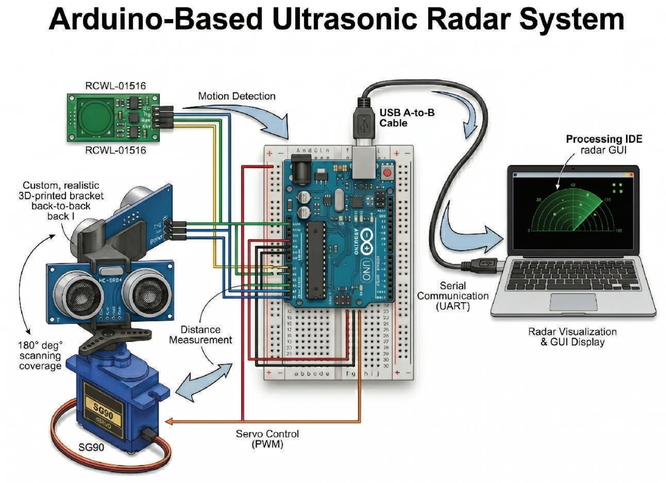
WORKING PRINCIPLE & METHODOLOGY



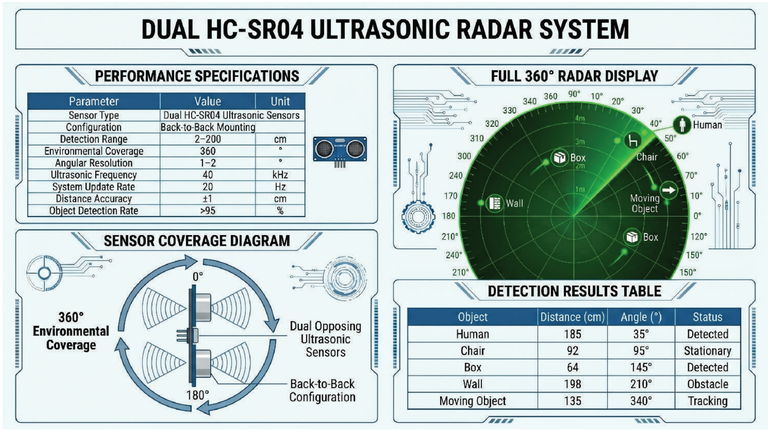
FLOWCHART



SYSTEM CONNECTION DIAGRAM



RESULTS & PERFORMANCE



CONCLUSION & FUTURE WORK

The successful implementation of this project demonstrates that an Arduino-based ultrasonic radar can serve as an efficient solution for real-time mapping, object tracking, and distance measurement within a short-range environment. The system successfully calculated target coordinates, including both angular positioning and exact physical distance.

Future Developments:
Machine Learning Classification
Wireless Communication
Multi-Sensor Fusion
IoT Integration

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